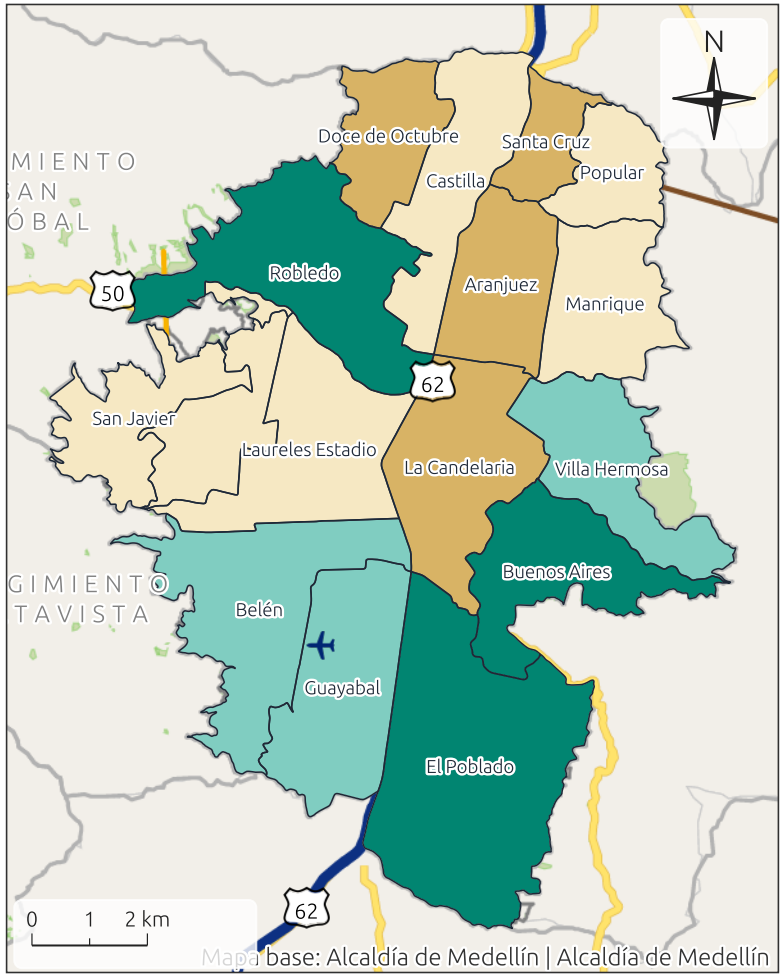
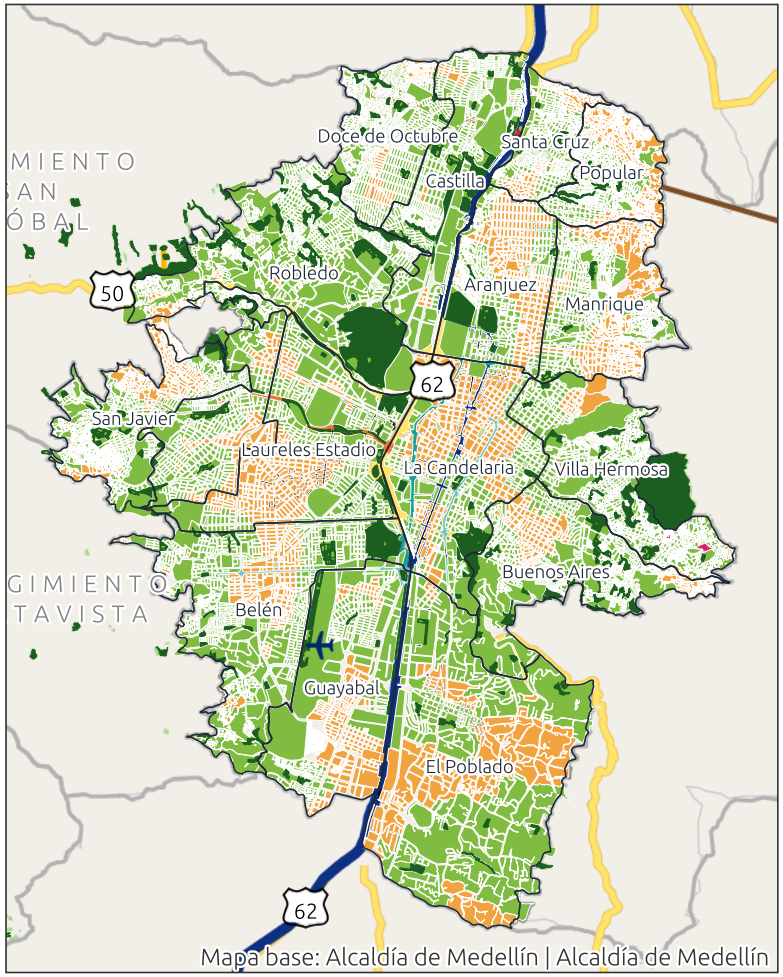
The 3-30-300 utopia in Medellín

A green urban planning rule (Konijnendijk, 2021): 3 trees in view from every home, 30 % tree canopy in every neighbourhood and a park within 300 m.
Public data can measure two of them: 6 of the 16 comunas pass 30 % vegetation, and 7 in 10 city blocks have a park within 300 m.

30 · Vegetation by comuna (NDVI from the 2024 orthophoto)



300 · Each city block and its nearest park of at least 0.5 ha



And the 3?

Seeing 3 trees from every home can't be measured with public data: Medellín's tree inventory records the work done on trees, not every tree, and what you see from a window needs street-level imagery.

By comuna

Above 30 %: El Poblado (52 %), Buenos Aires (43 %), Robledo (41 %), Belén (34 %), Villa Hermosa (34 %) and Guayabal (31 %).
The least green: Santa Cruz and Doce de Octubre, at 13 % each.

Reading it

NDVI counts all vegetation: trees, lawns and gardens. Guayabal adds the grass of Olaya Herrera airport. It approximates the canopy; it isn't the canopy.
The 300 m are a straight line, edge to edge, not a walk.

Vegetación · Vegetation
- < 20 %
- 20–30 %
- 30–40 %
- ≥ 40 %

Parques ≥ 0,5 ha · Parks ≥ 0.5 ha
Manzanas · Blocks
- ≤ 300 m
- > 300 m

How it was made: all in RustGIS web (web.rustgis.app), with four scripts. 1 · Data: comunas, parks, city blocks and a vegetation image (NDVI > 0.3) that the city's server computes from the 2024 orthophoto.
2 · View: Zonal Statistics (each comuna's mean is its vegetation share), Merge, Simplify and Near (distance from each block to the nearest park). 3–4 · Layouts. Scripts: github.com/jgregoriorincon/30DayMapChallenge-2026

Data: Alcaldía de Medellín, Administrative Department of Planning (GeoMedellín), CC BY-SA 4.0: November 2024 orthophoto, city blocks, existing public space and comunas;
accessed 2026-11-07. Basemap: Alcaldía de Medellín. This map and its project: CC BY-SA 4.0.